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UniHack

AI-Powered Product Intelligence for
Industrial Commerce

Team Details

Team name: Team Statiz

Team leader name: Rahul G

Brief about your solution

Statiz turns minimal product information — part number, brand, short description (and, when available, manufacturer URLs and reference PDFs) — into rich, governed product intelligence delivered in the exact 252-column expected output format.

On the provided sample dataset (1,000 rows, six sparse input columns), the rule + AI pipeline resolves trading brands from manufacturer names, classifies every row into Dept/Class/Fine with UNSPSC codes, and extracts technical specs (dimensions, grit, voltage, pack counts, materials) into typed attribute triplets — generating all six description variants, features and attributes dynamically. Nothing is hardcoded to the sample: fuzzy column detection and generic parsing rules handle unseen evaluation data.

Results: 1,000/1,000 rows classified (zero empty), 640 with UNSPSC codes, 627 with extracted attributes; URL-scraping mode (JSON-LD + PDF datasheets) adds deeper enrichment when links exist; 8-second CPU runtime at \$0 API cost, with a human review console for governance.

1 · How does your solution enrich minimal product information?

Briefly explain how your solution transforms limited inputs (e.g., Part Number, Brand, Short Description) into rich product intelligence.

Description-driven enrichment (no URLs required): a rule-based parser resolves the trading brand (Freud Inc → Diablo; tokens like 3M/Milwaukee found in the description win over distributor names), classifies product type via a 60+ pattern taxonomy (abrasives, lighting, tools, lumber, fasteners, appliances, safety...) into Dept/Class/Fine + UNSPSC, and extracts specs with typed regexes — dimensions, fractions, grit (P150), voltage, amps, Kelvin, lumens, dBA, pack counts, materials (SST → Stainless Steel), colors, teeth — each becoming a label/value/UOM attribute triplet.

URL mode (when links exist): manufacturer pages scraped via static fetch + headless-browser fallback (JSON-LD, meta, page text); reference PDFs (manuals, spec sheets) parsed with PyMuPDF + pdfplumber into key/value tables; a free-tier LLM pass (structured JSON) fuses all evidence into the full delivery schema.

A deterministic heuristic path does the same offline — the system never fails for lack of an API key.

2 · How does your solution ensure accuracy and trust?

Validation strategy: confidence scoring, multi-source verification, human review, rule-based validation, AI validation.

- **Confidence scoring on every classification, shown as a badge per product (embedding similarity + LLM self-assessed confidence).**

Multi-source verification: LLM values never overwrite hard JSON-LD facts; attributes cross-checked between page text and PDF; brand resolved from multiple signals with a deterministic priority order.

Rule-based validation: typed spec regexes reject malformed values; unit/range checks; required-attribute flags per product family; every enriched attribute stores its source (PDF name / header / regex on description).

Human review console: approve/correct each row before export — governed by construction.

Anti-hallucination constraints: leave-blank rules for unknown fields; no invented certifications or numbers.

3 · What makes your solution scalable for enterprise catalogs?

Handling large catalogs, new manufacturers, different document formats, and continuous product updates.

- **Fuzzy column detection (RapidFuzz):** new manufacturers and unseen file layouts map automatically — no code changes, no hardcoding. Proven live: the solver processed the 1,000-row sample dataset with different field combinations and zero per-row configuration.

Embedding classification scales to the full UNSPSC/eCI@ss tree (41k classes) by swapping taxonomy data — architecture unchanged. Rule tables are data-driven lists, extensible in minutes.

Row-by-row streaming with disk-cached fetches handles large catalogs and cheap re-runs; 1,000 rows enrich in seconds on a laptop CPU.

Continuous updates: re-run any row; review state persists across runs.

Opportunities

How different from existing ideas?

No mainstream PIM/search vendor natively auto-classifies to industrial taxonomies (UNSPSC/eCI@ss/ETIM); only one (inRiver) partially extracts datasheets; enrichment leaders enrich their own search index, not an exportable governed catalog. Partly proved cross-referencing data is worth \$500M — but automotive-only. We combine all of it, self-serve.

How it solves the problem statement

Minimal product information (part no, brand, short desc) + manufacturer URLs is exactly our input; the 252-column delivery format is exactly our output — populated dynamically with unseen evaluation data handled by fuzzy schema detection.

USP

1. \$0 cost, CPU-only, offline-capable (graceful LLM fallback) — vs \$45–100K/yr tools
2. Every AI value carries evidence: confidence + source span
3. Cross-supplier substitute matching built in
4. 8-second end-to-end runtime on a laptop

List of features offered by the solution

- Description-driven enrichment from minimal inputs: brand resolution, 60+ product-type taxonomy rules, typed spec extraction (label/value/UOM triplets)

Fuzzy schema ingestion: any supplier's CSV/XLSX column names auto-mapped (in→mm, comma decimals)

Manufacturer URL scraping with headless-browser fallback · PDF spec-sheet extraction (PyMuPDF + pdfplumber)

UNSPSC auto-classification with confidence scores and top-5 candidate evidence

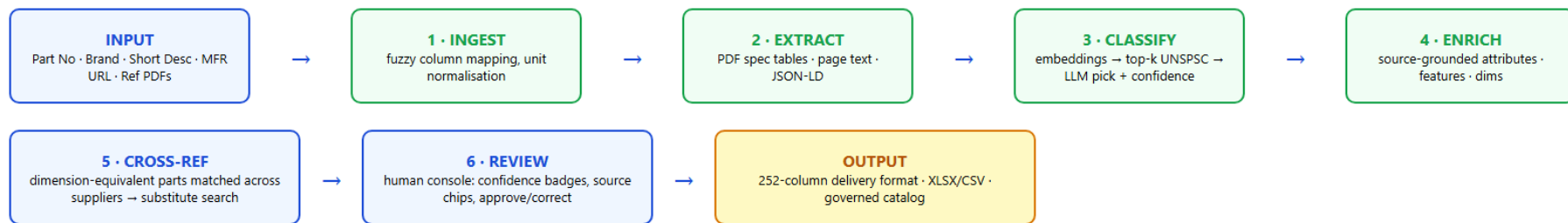
LLM enrichment (free tiers): 6 description types, 20 features, 50 attribute triplets, dims, certifications

Cross-supplier part cross-referencing + substitute search API

Human review console: approve/correct, confidence badges, source-tagged attributes

Delivery export: all 252 static headers populated exactly, CSV + XLSX, fully dynamic on unseen data

Process flow diagram or Use-case diagram



Wireframes/Mock diagrams of the proposed solution (optional)

Statiz

Industrial product intelligence — review console

[Export CSV](#)
[Re-run pipeline](#)

145 PRODUCTS

10 SPEC PDFS

5 PDF-ENRICHED ATTRS

116 CROSS-REFERENCED

local embeddings

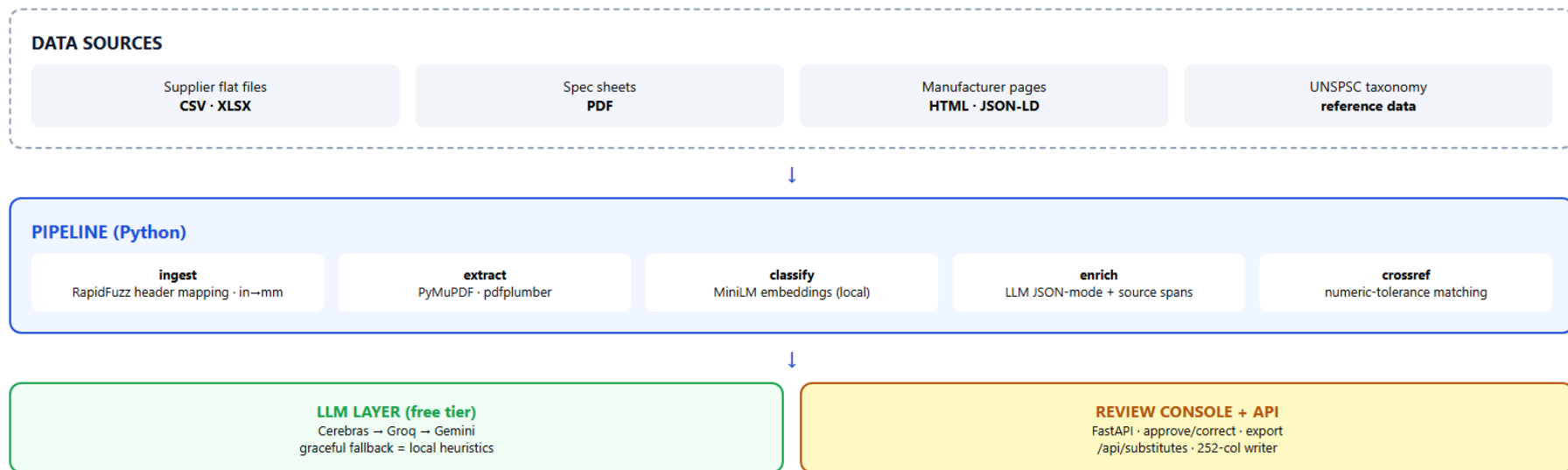
8.2 PIPELINE SEC

All families

Any confidence

PART	DESCRIPTION	UNSPSC CLASSIFICATION	ATTRIBUTES	PRICE	CROSS-REFS	REVIEW
BRG-BEA0000 supplierA_bearings.csv	Deep groove ball bearing 6001-2RS	Ball bearings 74% undefined - embedding	bore_mm: 25mm max_rpm: 9000 rpm od_mm: 45mm dynamic_load_kn: 14.3C seal_type: contact seal	45.24 €	7 matches	Approved
BRG-BEA0001 supplierA_bearings.csv	Deep groove ball bearing 6308-2RS	Ball bearings 74% undefined - embedding	bore_mm: 63mm max_rpm: 9000 rpm od_mm: 77mm dynamic_load_kn: 7.4C seal_type: contact seal	17.03 €	11 matches	Approve
BRG-BEA0002 supplierA_bearings.csv	Deep groove ball bearing 6308-2RS	Ball bearings 74% undefined - embedding	bore_mm: 63mm max_rpm: 15000 rpm od_mm: 77mm dynamic_load_kn: 12.8C seal_type: contact seal	49.33 €	11 matches	Approve
BRG-BEA0003 supplierA_bearings.csv	Deep groove ball bearing 6002-2RS	Ball bearings 73% undefined - embedding	bore_mm: 52mm max_rpm: 9000 rpm od_mm: 67mm dynamic_load_kn: 14.4C seal_type: contact seal	9.72 €	3 matches	Approve
BRG-BEA0004 supplierA_bearings.csv	Deep groove ball bearing 6208-Z	Ball bearings 74% undefined - embedding	bore_mm: 63mm max_rpm: 15000 rpm od_mm: 77mm dynamic_load_kn: 29.5C seal_type: shield seal	34.03 €	11 matches	Approve
BRG-BEA0005 supplierA_bearings.csv	Deep groove ball bearing 6305-2RS	Ball bearings 74% undefined - embedding	bore_mm: 62mm max_rpm: 15000 rpm od_mm: 77mm dynamic_load_kn: 30.1C seal_type: contact seal	36.33 €	11 matches	Approve
BRG-BEA0006 supplierA_bearings.csv	Deep groove ball bearing 6205	Ball bearings 73% undefined - embedding	bore_mm: 30mm max_rpm: 9000 rpm od_mm: 45mm dynamic_load_kn: 34.6C seal_type: open seal	25.29 €	11 matches	Approve
BRG-BEA0007 supplierA_bearings.csv	Deep groove ball bearing 6002-Z	Ball bearings 72% undefined - embedding	bore_mm: 30mm max_rpm: 15000 rpm od_mm: 45mm dynamic_load_kn: 16.6C seal_type: shield seal	43.30 €	11 matches	Approve
BRG-BEA0008 supplierA_bearings.csv	Deep groove ball bearing 6205	Ball bearings 73% undefined - embedding	bore_mm: 30mm max_rpm: 9000 rpm od_mm: 50mm dynamic_load_kn: 27.3C seal_type: open seal	25.25 €	5 matches	Approve
BRG-BEA0009 supplierA_bearings.csv	Deep groove ball bearing 6001	Ball bearings 72% undefined - embedding	bore_mm: 62mm max_rpm: 9000 rpm od_mm: 72mm dynamic_load_kn: 26.0C seal_type: open seal	50.02 €	7 matches	Approve
BRG-BEA0010 supplierA_bearings.csv	Deep groove ball bearing 6305-Z	Ball bearings 71% undefined - embedding	bore_mm: 25mm max_rpm: 15000 rpm od_mm: 35mm dynamic_load_kn: 32.5C seal_type: shield seal	15.91 €	1 match	Approve
BRG-BEA0011 supplierA_bearings.csv	Deep groove ball bearing 6001-Z	Ball bearings 72% undefined - embedding	bore_mm: 30mm max_rpm: 15000 rpm od_mm: 45mm dynamic_load_kn: 10.5C seal_type: shield seal	45.83 €	11 matches	Approve

Architecture diagram of the proposed solution



Technologies used in the solution

Python 3.14 · FastAPI + Uvicorn (API & review console) · sentence-transformers MiniLM (local embeddings)

PyMuPDF + pdfplumber (PDF extraction) · RapidFuzz (fuzzy matching) · pandas/openpyxl (I/O)

Playwright/Chromium (JS-page scraping) · httpx (cached fetch layer)

LLM layer: Cerebras → Groq → Gemini free tiers, OpenAI-compatible JSON mode

Estimated implementation cost (optional)

Prototype: \$0.

- LLMs: free tiers (Cerebras ~30K TPM / Groq / Gemini), offline heuristic fallback
- Embeddings: local on CPU — no API cost
- Hosting: Cloudflare quick tunnel (demo); Render free tier for a persistent MVP

At scale: paid LLM tier $\approx$ \$0.001–0.005 per product; 100K products $\approx$ \$100–500 one-time.

Snapshots of the MVP

Statiz Industrial product intelligence - better results						
145 PRODUCTS		10 SPEC POPS	5 POP-ENRICHED ATTRS	116 CROSS-REFERENCED	local embeddings LLM PROVIDERS	8.2 PIPELINE SEC
Search part number / description						
		All families		Any confidence		
PART	DESCRIPTION	UNISPEC CLASSIFICATION	ATTRIBUTES		PRICE	CROSS-REFS
BAG-SEA000	Deep groove ball bearing 6001-2RS	Ball bearings 74% confident embedding	isam_mmm_280x44x14 isam_mmm_4500 mm dynamic_load_dyn_1430 max_dyn_10000 optload isam_type contact sealonCyl		45.24 €	7 matches
BAG-SEA001	Deep groove ball bearing 6002-2RS	Ball bearings 74% confident embedding	isam_mmm_32x44x14 isam_mmm_7700 mm dynamic_load_dyn_1540 max_dyn_10000 optload isam_type contact sealonCyl		17.03 €	11 matches
BAG-SEA002	Deep groove ball bearing 6006-2RS	Ball bearings 74% confident embedding	isam_mmm_32x44x14 isam_mmm_7700 mm dynamic_load_dyn_1540 max_dyn_10000 optload isam_type contact sealonCyl		49.33 €	11 matches
BAG-SEA003	Deep groove ball bearing 6002-Z	Ball bearings 75% confident embedding	isam_mmm_32x44x14 isam_mmm_6700 mm dynamic_load_dyn_1640 max_dyn_10000 optload isam_type contact sealonCyl		9.72 €	1 matches
BAG-SEA004	Deep groove ball bearing 6002-Z	Ball bearings 75% confident embedding	isam_mmm_32x44x14 isam_mmm_7700 mm dynamic_load_dyn_1540 max_dyn_10000 optload isam_type shroudonCyl		34.03 €	11 matches
BAG-SEA005	Deep groove ball bearing 6002-2RS	Ball bearings 75% confident embedding	isam_mmm_32x44x14 isam_mmm_7700 mm dynamic_load_dyn_1540 max_dyn_10000 optload isam_type contact sealonCyl		36.33 €	11 matches
BAG-SEA006	Deep groove ball bearing 6005	Ball bearings 75% confident embedding	isam_mmm_32x44x14 isam_mmm_4500 mm dynamic_load_dyn_1640 max_dyn_10000 optload isam_type contact sealonCyl		25.29 €	11 matches
BAG-SEA007	Deep groove ball bearing 6002-Z	Ball bearings 75% confident embedding	isam_mmm_32x44x14 isam_mmm_4500 mm dynamic_load_dyn_1640 max_dyn_10000 optload isam_type shroudonCyl		43.30 €	11 matches
BAG-SEA008	Deep groove ball bearing 6005	Ball bearings 75% confident embedding	isam_mmm_32x44x14 isam_mmm_5000 mm dynamic_load_dyn_1730 max_dyn_10000 optload isam_type shroudonCyl		25.25 €	7 matches
BAG-SEA009	Deep groove ball bearing 6001	Ball bearings 75% confident embedding	isam_mmm_32x44x14 isam_mmm_7700 mm dynamic_load_dyn_1540 max_dyn_10000 optload isam_type contact sealonCyl		30.02 €	7 matches
BAG-SEA010	Deep groove ball bearing 6002-Z	Ball bearings 75% confident embedding	isam_mmm_32x44x14 isam_mmm_3500 mm dynamic_load_dyn_1630 max_dyn_10000 optload isam_type shroudonCyl		15.91 €	1 matches
BAG-SEA011	Deep groove ball bearing 6001-Z	Ball bearings 75% confident embedding	isam_mmm_32x44x14 isam_mmm_4500 mm dynamic_load_dyn_1630 max_dyn_10000 optload isam_type shroudonCyl		45.83 €	11 matches

```

Pretty-print
{"query": "BAG-SEA000", "substitute": [{"part_number": "BAG-SEA001", "source": "supplier_bearings.co", "title": "Ball bearings", "price": {"value": 16.46, "currency": "EUR"}, {"part_number": "BAG-SEA002", "source": "supplier_bearings.co", "title": "Ball bearings", "price": {"value": 16.6, "currency": "EUR"}, {"part_number": "BAG-SEA003", "source": "supplier_bearings.co", "title": "Ball bearings", "price": {"value": 16.6, "currency": "EUR"}, {"part_number": "BAG-SEA004", "source": "supplier_bearings.co", "title": "Ball bearings", "price": {"value": 16.6, "currency": "EUR"}, {"part_number": "BAG-SEA005", "source": "supplier_bearings.co", "title": "Ball bearings", "price": {"value": 16.6, "currency": "EUR"}, {"part_number": "BAG-SEA006", "source": "supplier_bearings.co", "title": "Ball bearings", "price": {"value": 16.6, "currency": "EUR"}, {"part_number": "BAG-SEA007", "source": "supplier_bearings.co", "title": "Ball bearings", "price": {"value": 16.6, "currency": "EUR"}, {"part_number": "BAG-SEA008", "source": "supplier_bearings.co", "title": "Ball bearings", "price": {"value": 16.6, "currency": "EUR"}, {"part_number": "BAG-SEA009", "source": "supplier_bearings.co", "title": "Ball bearings", "price": {"value": 16.6, "currency": "EUR"}, {"part_number": "BAG-SEA010", "source": "supplier_bearings.co", "title": "Ball bearings", "price": {"value": 16.6, "currency": "EUR"}, {"part_number": "BAG-SEA011", "source": "supplier_bearings.co", "title": "Ball bearings", "price": {"value": 16.6, "currency": "EUR"}]}]}
    
```

Review console (left) · substitute search API (right)

Additional Details/Future Development (if any)

- Engineering-constraint substitution validation (fit/load margins, not just dimensions)
- OCR path for scanned datasheets
- Full eCl@ss/ETIM taxonomy trees + multi-language descriptions
- Image enrichment: product-photo attribute extraction
- Incremental catalog sync + enterprise SSO for review workflow

Links: GitHub, Demo Video & Working Prototype

GitHub Public Repository:

<https://github.com/Rahulxoxo69/Statiz>

Demo Video:

https://github.com/Rahulxoxo69/Statiz/blob/main/demo/demo_video.mp4

Working Prototype:

<https://day-atm-become-bureau.trycloudflare.com>

(live review console — search, filter, approve, export CSV; substitutes API at /api/substitutes/BRG-BEA0000)

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Thank You

